

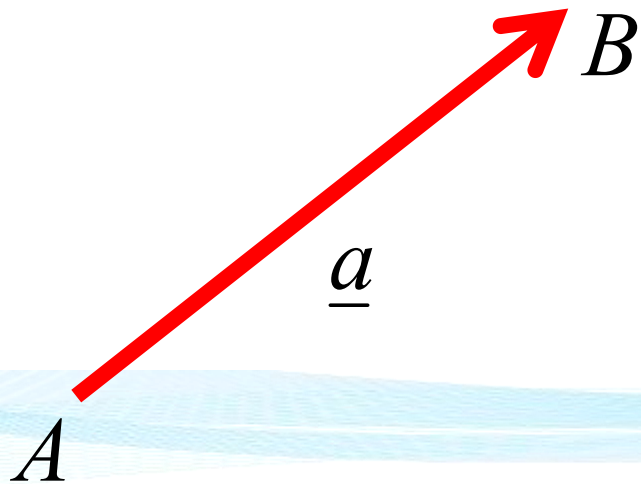
Chp17 平面向量

Chp17.1 向量(Vector)

向量(或称“有向线段”):有大小、有方向的量

Pg. 92

(如: 力、位移、速度、加速度...)



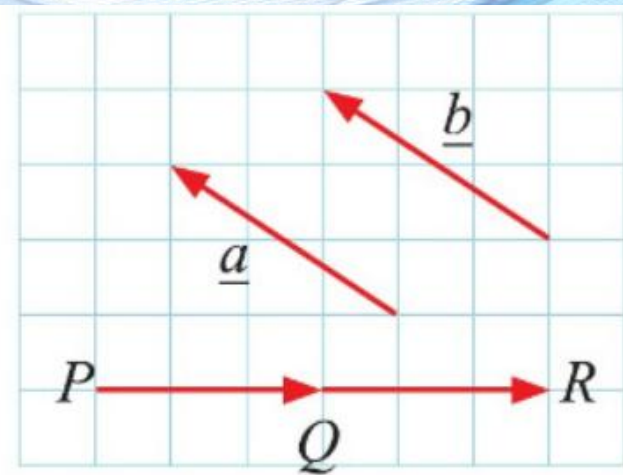
如左图, 有向线段**AB**记作: $\overrightarrow{AB}$

亦可表示成 $\underline{a}, \underset{\sim}{a}, \overline{a}, \vec{a}$ 。

相等向量equal vector:

大小相等、方向相同的向量

(如图 $\underline{a} = \underline{b}, \overrightarrow{PQ} = \overrightarrow{QR}$)



a) 如右图所示, $ABCDEF$ 为一正六边形,

已知 $\overrightarrow{OA} = \underline{a}, \overrightarrow{OB} = \underline{b}, \overrightarrow{AB} = \underline{c}$, 求以 $\underline{a}, \underline{b}, \underline{c}$ 表示

$$\overrightarrow{DO} = \underline{a}$$

$$\overrightarrow{EO} = \underline{b}$$

$$\overrightarrow{ED} = \underline{c}$$

$$\overrightarrow{FO} =$$

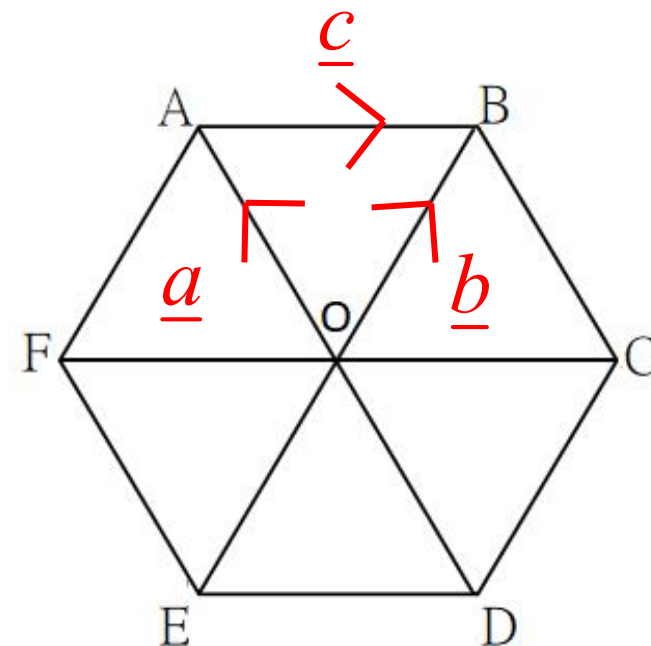
$$\overrightarrow{OC} =$$

$$\overrightarrow{CB} =$$

$$\overrightarrow{DC} =$$

$$\overrightarrow{EF} =$$

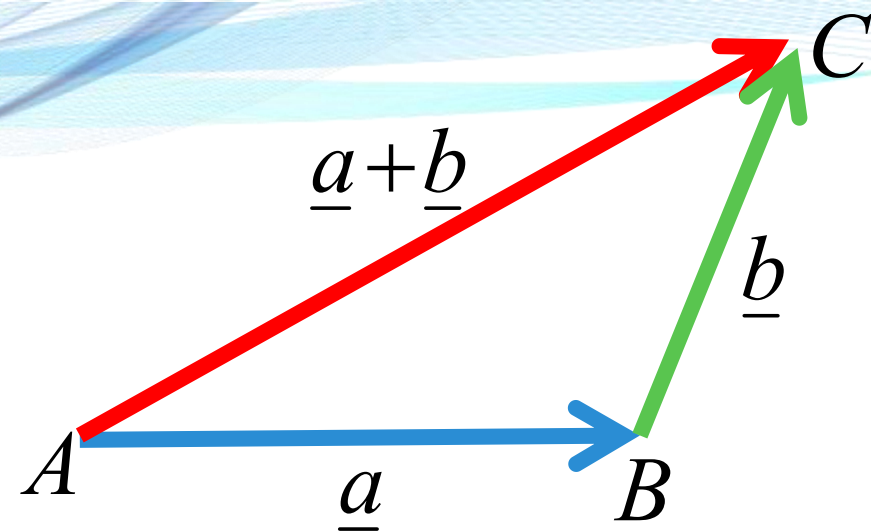
$$\overrightarrow{FA} =$$



向量的加法：加法绘图：首尾相连

(一) 向量求和三角形法则

(triangle law of addition) 常用



已知 $\overrightarrow{AB} = \underline{a}$, $\overrightarrow{BC} = \underline{b}$, 则 $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = \underline{a} + \underline{b}$

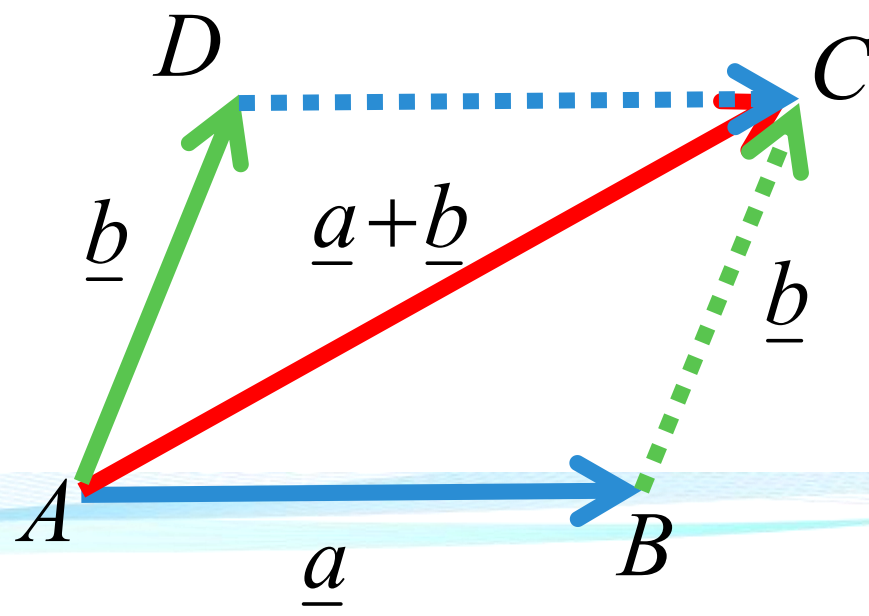
$\overrightarrow{\text{起}}\overrightarrow{\text{终}} = \overrightarrow{\text{起}}X + X\overrightarrow{\text{终}}$,
其中X为任一点

(二) 向量求和四边形法则

(parallelogram law of addition)

已知 $\overrightarrow{AB} = \underline{a}$, $\overrightarrow{AD} = \underline{b}$, 则

$$\overrightarrow{AB} + \overrightarrow{AD} = \overrightarrow{AB} + \overrightarrow{BC} = \underline{a} + \underline{b} = \overrightarrow{AC}$$



向量加法性质:

$$i) \underline{a} + \underline{b} = \underline{b} + \underline{a} \quad (\text{加法交换律})$$

$$ii) (\underline{a} + \underline{b}) + \underline{c} = \underline{a} + (\underline{b} + \underline{c}) \quad (\text{加法结合律})$$

a) Diagram shows a pentagon $PQRST$ with marked vectors.

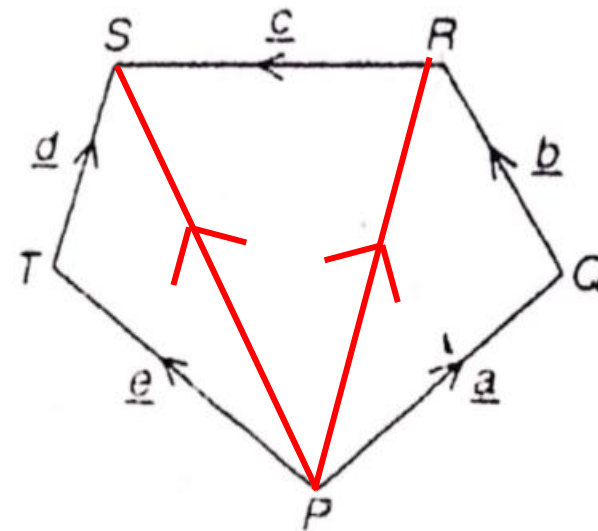
$$i) \overrightarrow{PR}$$

$$ii) \overrightarrow{PS}$$

$$\begin{aligned} i) \overrightarrow{PR} &= \overrightarrow{PQ} + \overrightarrow{QR} \\ &= \underline{a} + \underline{b} \end{aligned}$$

$$\begin{aligned} ii) \overrightarrow{PS} &= \overrightarrow{PT} + \overrightarrow{TS} \\ &= \underline{e} + \underline{d} \end{aligned}$$

$$\begin{aligned} \text{或 } \overrightarrow{PS} &= \overrightarrow{PR} + \overrightarrow{RS} \\ &= \underline{a} + \underline{b} + \underline{c} \end{aligned}$$



Diagram

零向量zero vector (记作: $\underline{0}$)

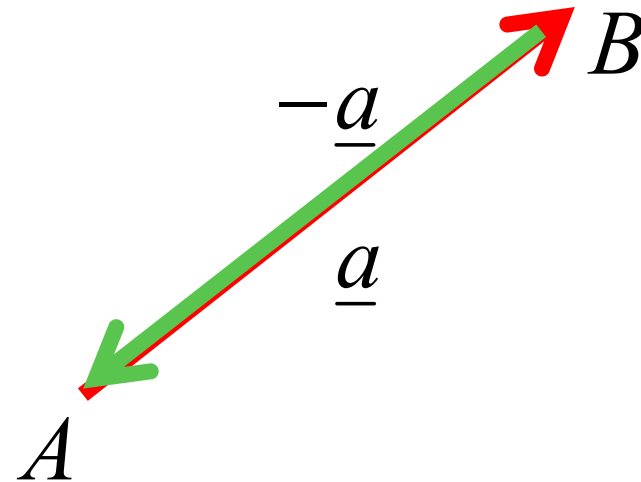
在 $\overrightarrow{AB}$ 中, 若B与A重合, 即 $\overrightarrow{AA} = \underline{0}$ (没有大小、方向不明)

$$\underline{a} + \underline{0} = \underline{a} \quad \text{或} \quad \overrightarrow{AB} + \overrightarrow{BB} = \overrightarrow{AB}$$

逆向量inverse vector

当A与B不重合时, $\overrightarrow{AB}$ 与 $\overrightarrow{BA}$ 是不同向量, 即 $\overrightarrow{AB} \neq \overrightarrow{BA}$

且 $\overrightarrow{AB} + \overrightarrow{BA} = \overrightarrow{AA} = \underline{0}$, 则 $\overrightarrow{BA}$ 为 $\overrightarrow{AB}$ 的逆向量, $\overrightarrow{BA}$ 记作 $-\overrightarrow{AB}$
(大小相等、方向相反)



向量的减法:

利用逆向量, 将向量相减定义为: 设 $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$

$$\underline{a} - \underline{b} = \underline{a} + (-\underline{b})$$

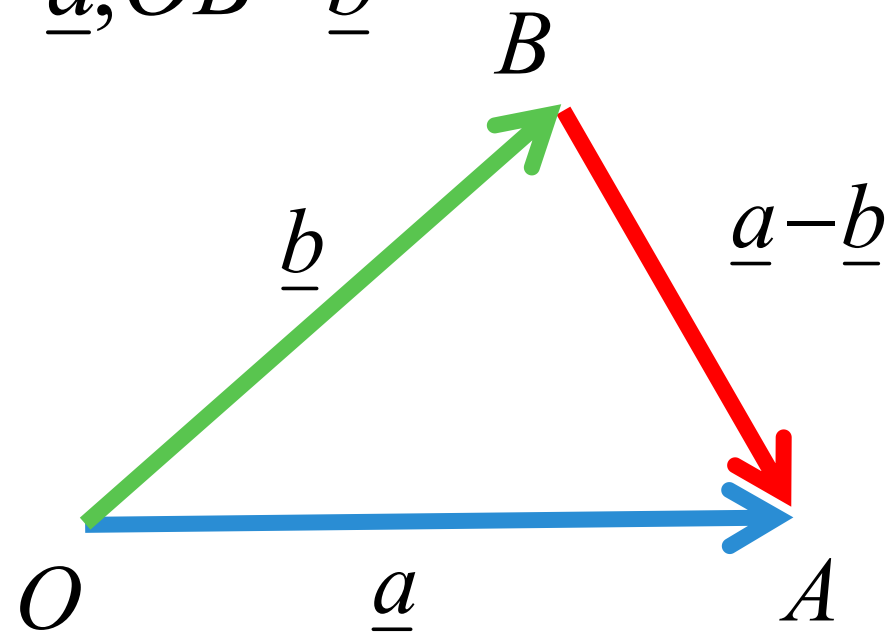
$$= \overrightarrow{OA} - \overrightarrow{OB}$$

$$= \overrightarrow{OA} + (-\overrightarrow{OB})$$

$$= \overrightarrow{OA} + \overrightarrow{BO}$$

$$= \overrightarrow{BO} + \overrightarrow{OA}$$

$$= \overrightarrow{BA}$$



$$\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB}$$

$\overrightarrow{起终} = \overrightarrow{X终} - \overrightarrow{X起}$, 其中 X 为任一点

b) 如图所示, $ABCDEF$ 为一正六边形,

已知 $\overrightarrow{AB} = \underline{a}$, $\overrightarrow{BC} = \underline{b}$, 求以 $\underline{a}, \underline{b}$ 表示,

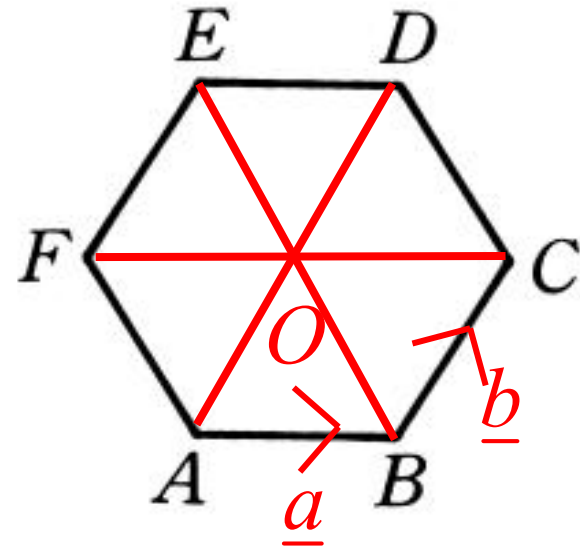
i) $\overrightarrow{AC}$

ii) $\overrightarrow{ED}$

iii) $\overrightarrow{EF}$

iv) $\overrightarrow{CD}$

v) $\overrightarrow{FA}$



$$\begin{aligned} i) \overrightarrow{AC} &= \overrightarrow{AB} + \overrightarrow{BC} \\ &= \underline{a} + \underline{b} \end{aligned}$$

$$\begin{aligned} iii) \overrightarrow{EF} &= \overrightarrow{CB} \\ &= -\overrightarrow{BC} \\ &= -\underline{b} \end{aligned}$$

$$\begin{aligned} iii) \overrightarrow{CD} &= \overrightarrow{BO} \\ &= \overrightarrow{BA} + \overrightarrow{AO} \\ &= -\overrightarrow{AB} + \overrightarrow{BC} \\ &= -\underline{a} + \underline{b} \end{aligned}$$

$$\begin{aligned} iv) \overrightarrow{FA} &= \overrightarrow{DC} \\ &= -\overrightarrow{CD} \\ &= \underline{a} - \underline{b} \end{aligned}$$

$$\begin{aligned} ii) \overrightarrow{ED} &= \overrightarrow{AB} \\ &= \underline{a} \end{aligned}$$

根据右图，填空：

$$(a) \overrightarrow{DB} + \overrightarrow{BE} =$$

$$(b) \overrightarrow{AD} + \overrightarrow{DB} + \overrightarrow{BC} =$$

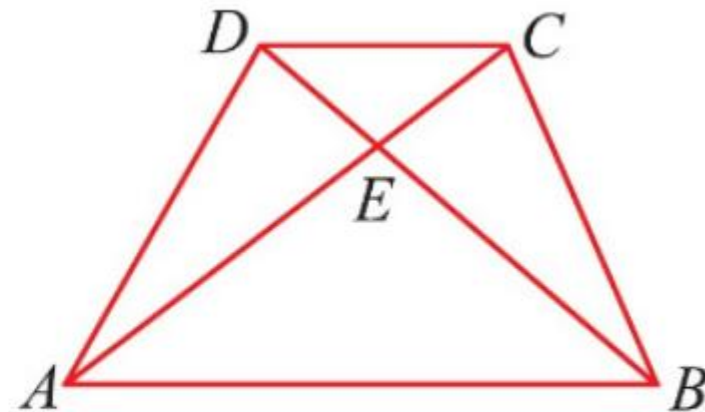
$$(c) \overrightarrow{AE} + \quad = \overrightarrow{AB}$$

$$(d) \overrightarrow{AD} + \quad + \overrightarrow{EC} = \overrightarrow{AC}$$

$$(e) \overrightarrow{AD} - \overrightarrow{AE} =$$

$$(f) \overrightarrow{BC} - \overrightarrow{AC} =$$

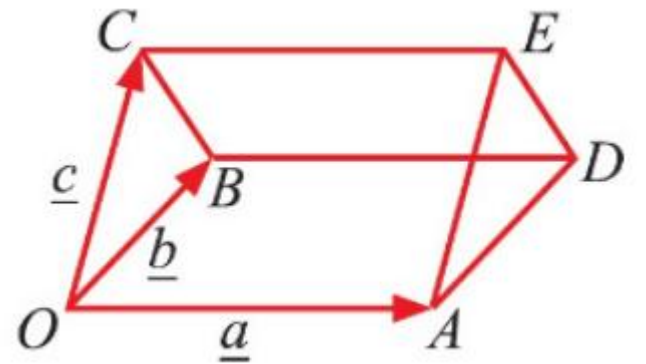
$$(g) \overrightarrow{AE} - \overrightarrow{BE} - \overrightarrow{AD} =$$



第3题用图

右图中, $OADB$ 与 $OAEC$ 是平行四边形, $\overrightarrow{OA} = \underline{a}, \overrightarrow{OB} = \underline{b}, \overrightarrow{OC} = \underline{c}$ 。

以 $\underline{a}, \underline{b}, \underline{c}$ 表示 $\overrightarrow{AD}, \overrightarrow{BC}, \overrightarrow{ED}, \overrightarrow{OD}$ 及 $\overrightarrow{OE}$ 。



根据右图，填空：

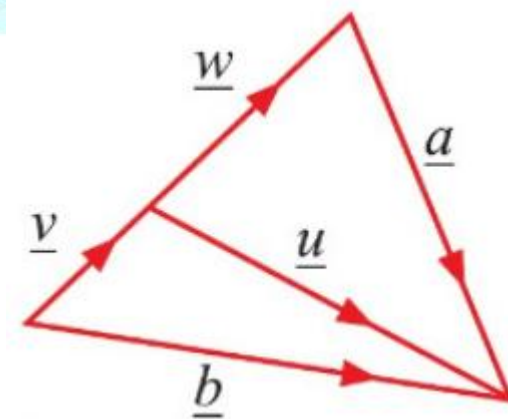
(a) 化简 $\underline{a} - \underline{u}$

(b) 化简 $\underline{v} + \underline{u} - \underline{b}$

(c) 求 $\underline{x}$ 使得 $\underline{x} + \underline{w} = \underline{u}$

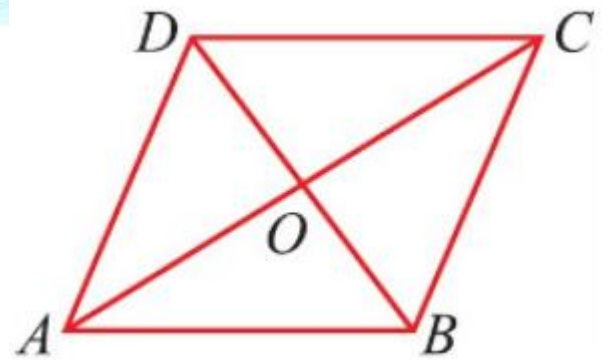
(d) 求 $\underline{x}$ 使得 $\underline{x} - \underline{a} = \underline{v} + \underline{w}$

(e) 求 $\underline{x}$ 使得 $\underline{w} - \underline{x} = \underline{b} - \underline{a}$



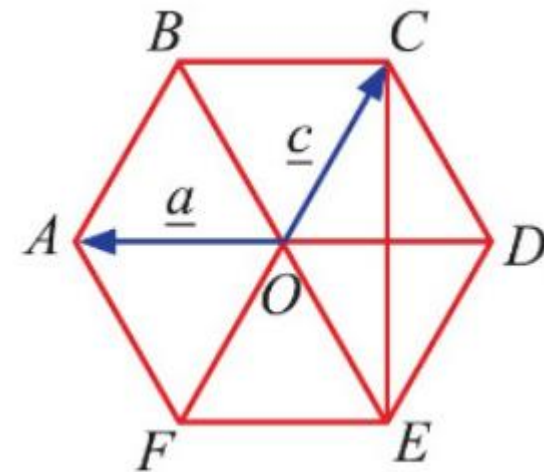
右图中, $ABCD$ 是平行四边形, $\overrightarrow{AB} = \underline{a}, \overrightarrow{DB} = \underline{b}$ 。

以 $\underline{a}, \underline{b}$ 表示 $\overrightarrow{AD}, \overrightarrow{AC}$ 及 $\overrightarrow{OA}$ 。

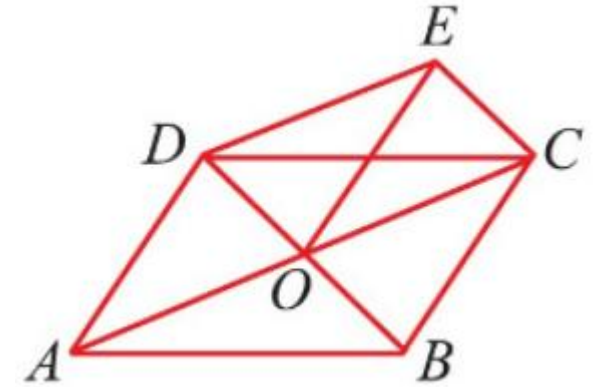


右图中, $ABCDEF$ 是正六边形, $\overrightarrow{OA} = \underline{a}, \overrightarrow{OC} = \underline{c}$ 。

以 $\underline{a}, \underline{c}$ 表示 $\overrightarrow{DE}, \overrightarrow{AD}, \overrightarrow{AF}, \overrightarrow{BE}$ 及 $\overrightarrow{CE}$ 。



右图中, $ABCD, OCED$ 是平行四边形, O 是 AC 与 BD 的交点,
若 $\overrightarrow{AB} = \underline{a}, \overrightarrow{BO} = \underline{b}$, 以 $\underline{a}, \underline{b}$ 表示 $\overrightarrow{BC}$ 及 $\overrightarrow{OE}$ 。



向量的数乘(the scalar

Pg. 97

multiplication of a vector)

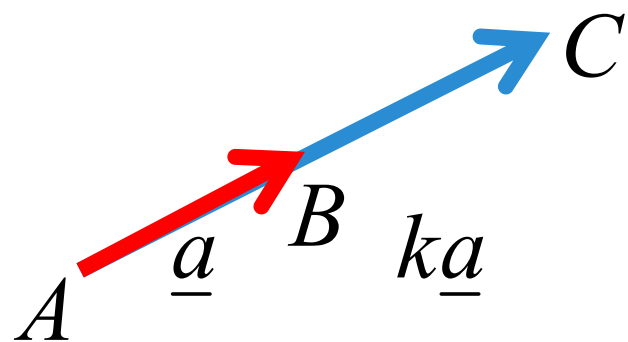
设 $\overrightarrow{AB} = \underline{a}$

若 $k = 0$,

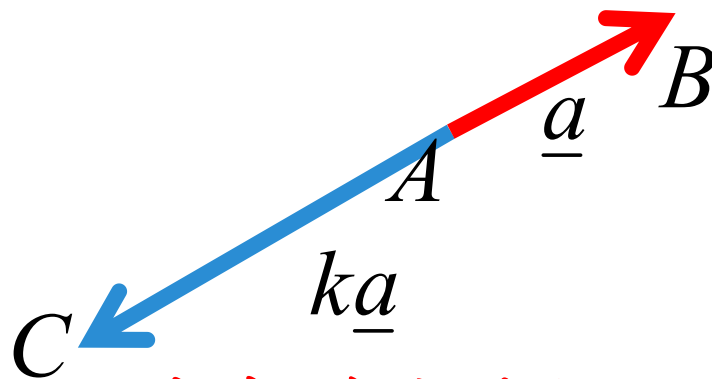
若 $k > 0$, $\overrightarrow{AC} = k\overrightarrow{AB} = k\underline{a}$

若 $k < 0$, $\overrightarrow{AC} = k\overrightarrow{AB} = k\underline{a}$

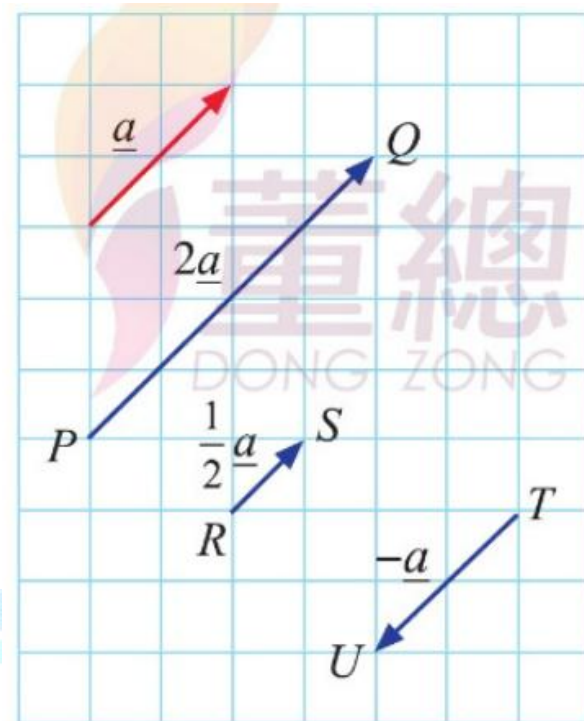
$\overrightarrow{AC} = k\overrightarrow{AB} = \underline{0}$



正向放大/缩小



反向放大/缩小



向量数乘性质:

$$i) k(l\underline{a}) = (kl)\underline{a} = kl\underline{a}$$

$$ii) (k+l)\underline{a} = k\underline{a} + l\underline{a}$$

$$iii) k(\underline{a} + \underline{b}) = k\underline{a} + k\underline{b}$$

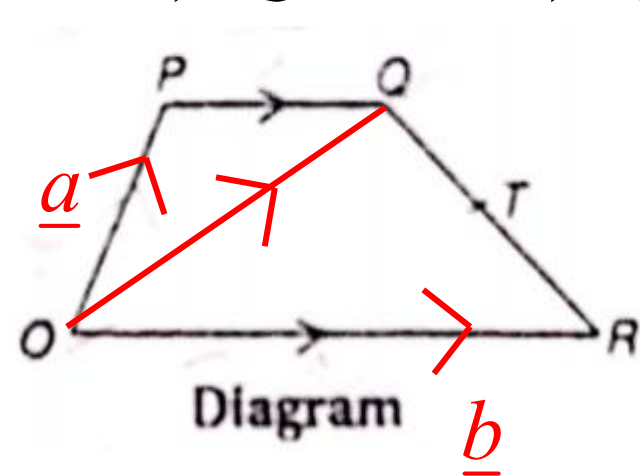
c) Diagram, $OPQR$ is a trapezium where OR is parallel to PQ

and $PQ = \frac{2}{3}OR$. Given that $\overrightarrow{OP} = \underline{a}$, $\overrightarrow{OR} = \underline{b}$, express in term of $\underline{a}, \underline{b}$,

i) $\overrightarrow{OQ}$

ii) $\overrightarrow{RQ}$

iii) $\overrightarrow{OT}$ given that T is the midpoint of QR .



$$i) \overrightarrow{OQ} = \overrightarrow{OP} + \overrightarrow{PQ}$$

$$= \overrightarrow{OP} + \frac{2}{3} \overrightarrow{OR}$$

$$= \underline{a} + \frac{2}{3} \underline{b}$$

$$ii) \overrightarrow{RQ} = \overrightarrow{RO} + \overrightarrow{OQ}$$

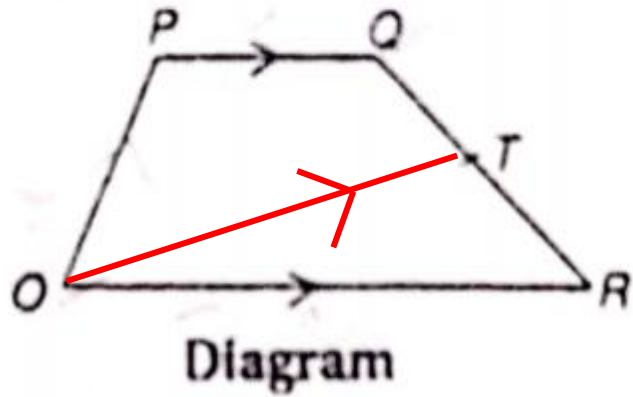
$$= -\underline{b} + \left(\underline{a} + \frac{2}{3} \underline{b} \right)$$

$$= \underline{a} - \frac{1}{3} \underline{b}$$

c) Diagram, $OPQR$ is a trapezium where OR is parallel to PQ

and $PQ = \frac{2}{3}OR$. Given that $\overrightarrow{OP} = \underline{a}$, $\overrightarrow{OR} = \underline{b}$, express in terms of $\underline{a}, \underline{b}$,

i) $\overrightarrow{OQ}$ ii) $\overrightarrow{RQ}$ iii) $\overrightarrow{OT}$ given that T is the midpoint of QR .



$$\text{iii) } \overrightarrow{OT} = \overrightarrow{OR} + \overrightarrow{RT}$$

$$= \overrightarrow{OR} + \frac{1}{2} \overrightarrow{RQ}$$

$$= \underline{b} + \frac{1}{2} \left(\underline{a} - \frac{1}{3} \underline{b} \right)$$

$$= \frac{1}{2} \underline{a} + \frac{5}{6} \underline{b}$$

$$\text{或 } \overrightarrow{OT} = \overrightarrow{OP} + \overrightarrow{PQ} + \overrightarrow{QT}$$

$$= \overrightarrow{OQ} - \frac{1}{2} \overrightarrow{RQ}$$

$$= \left(\underline{a} + \frac{2}{3} \underline{b} \right) - \frac{1}{2} \left(\underline{a} - \frac{1}{3} \underline{b} \right)$$

$$= \frac{1}{2} \underline{a} + \frac{5}{6} \underline{b}$$

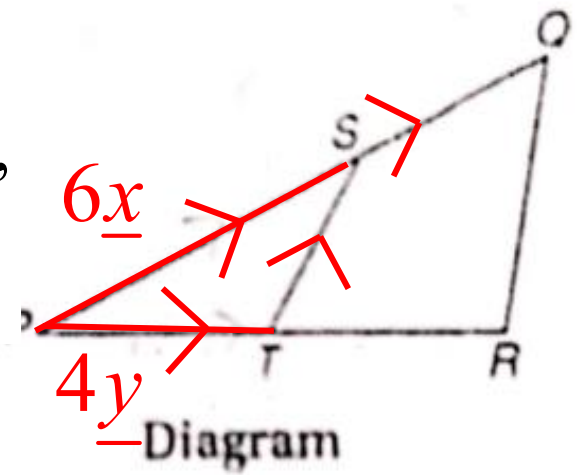
d) Diagram, T is midpoint of PR and $PS : SQ = 3 : 1$.

Given that $\overrightarrow{PS} = 6\underline{x}$, $\overrightarrow{PT} = 4\underline{y}$, express in term of $\underline{x}, \underline{y}$,

i) $\overrightarrow{TS}$

ii) $\overrightarrow{PQ}$

iii) $\overrightarrow{RS}$.



$$\begin{aligned} \text{i) } \overrightarrow{TS} &= \overrightarrow{TP} + \overrightarrow{PS} \\ &= -4\underline{y} + 6\underline{x} \\ &= 6\underline{x} - 4\underline{y} \end{aligned}$$

$$\because PS : SQ = 3 : 1$$

$$\therefore \frac{PS}{SQ} = 3$$

$$SQ = \frac{1}{3} PS$$

$$\text{ii) } \overrightarrow{PQ} = \overrightarrow{PS} + \overrightarrow{SQ}$$

$$= \overrightarrow{PS} + \frac{1}{3} \overrightarrow{PS}$$

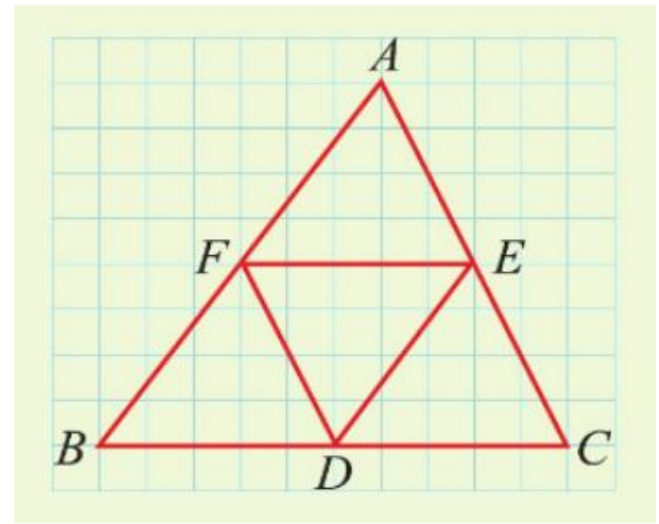
$$= \frac{4}{3} \overrightarrow{PS}$$

$$= \left(\frac{4}{3} \right) 6\underline{x}$$

$$= 8\underline{x}$$

$$\begin{aligned} \text{iii) } \overrightarrow{RS} &= \overrightarrow{RP} + \overrightarrow{PS} \\ &= 2\overrightarrow{TP} + \overrightarrow{PS} \\ &= 2(-4\underline{y}) + 6\underline{x} \\ &= 6\underline{x} - 8\underline{y} \end{aligned}$$

如图, $\overrightarrow{AB} = \underline{b}$, $\overrightarrow{AC} = \underline{c}$, 以为 $\underline{b}, \underline{c}$ 表示, $\overrightarrow{BC}, \overrightarrow{DF}, \overrightarrow{DE}, \overrightarrow{FE}$ 。



向量数乘的应用：

平行向量：

$$\overrightarrow{AB} // \overrightarrow{CD} \Leftrightarrow \text{存在非零常数 } k, \text{ 使得 } \overrightarrow{AB} = k \overrightarrow{CD}$$

三点共线：

$$A, B, C \text{ 三点共线} \Leftrightarrow \text{存在非零常数 } k, \text{ 使得 } \overrightarrow{AB} = k \overrightarrow{BC}, \text{ 且 } B \text{ 为公共点}$$

e) 如图, $\overrightarrow{AB} = \underline{c}$, $\overrightarrow{AC} = \underline{b}$, M, N 分别为 AB 和 AC 中点,

i) 求 $\overrightarrow{BC}$, $\overrightarrow{AM}$, $\overrightarrow{AN}$, $\overrightarrow{MN}$;

ii) $\overrightarrow{BC}$, $\overrightarrow{MN}$ 两者有什么关系?

$$\begin{aligned} i) \overrightarrow{BC} &= \overrightarrow{BA} + \overrightarrow{AC} \\ &= -\underline{c} + \underline{b} \end{aligned}$$

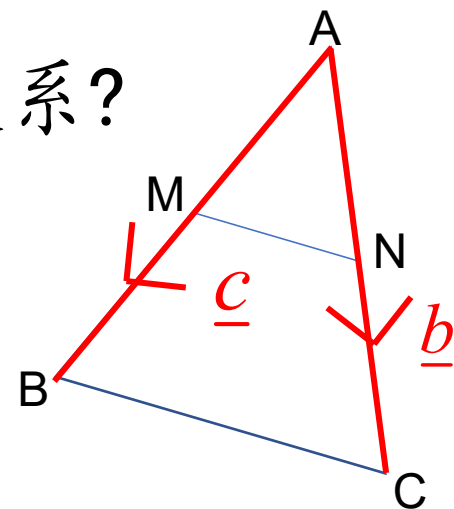
$$\begin{aligned} \overrightarrow{AM} &= \frac{1}{2} \overrightarrow{AB} \\ &= \frac{1}{2} \underline{c} \end{aligned}$$

$$\overrightarrow{AN} = \frac{1}{2} \overrightarrow{AC}$$

$$= \frac{1}{2} \underline{b}$$

$$\begin{aligned} \overrightarrow{MN} &= \overrightarrow{MA} + \overrightarrow{AN} \\ &= -\frac{1}{2} \underline{c} + \frac{1}{2} \underline{b} \end{aligned}$$

$$\begin{aligned} ii) \because \overrightarrow{BC} &= 2\overrightarrow{MN} \\ \therefore \overrightarrow{BC} &\parallel \overrightarrow{MN} \end{aligned}$$



f) 判断下列各向量是否平行，若是，则判断是否三点共线。

$$i) \overrightarrow{AB} = 2\underline{u}, \overrightarrow{CD} = 3\underline{u};$$

$$\therefore \overrightarrow{AB} = \frac{2}{3} \overrightarrow{CD} \quad \therefore \overrightarrow{AB} \parallel \overrightarrow{CD} \quad \text{三点不共线}$$

$$ii) \overrightarrow{PQ} = \underline{u}, \overrightarrow{RS} = -6\underline{v};$$

$$\therefore \overrightarrow{PQ} \neq \overrightarrow{RS} \quad \therefore \overrightarrow{PQ} \not\parallel \overrightarrow{RS}$$

$$iii) \overrightarrow{XY} = 1.5\underline{w}, \overrightarrow{YZ} = 7\underline{w};$$

$$\therefore \overrightarrow{XY} = \frac{1.5}{7} \overrightarrow{YZ} \quad \therefore \overrightarrow{XY} \parallel \overrightarrow{YZ}, \quad \text{三点共线, 且 } Y \text{ 为公共点}$$

$$iv) \overrightarrow{AB} = 3\underline{a} - \underline{b}, \overrightarrow{YZ} = 4.5\underline{a} - 1.5\underline{b}$$

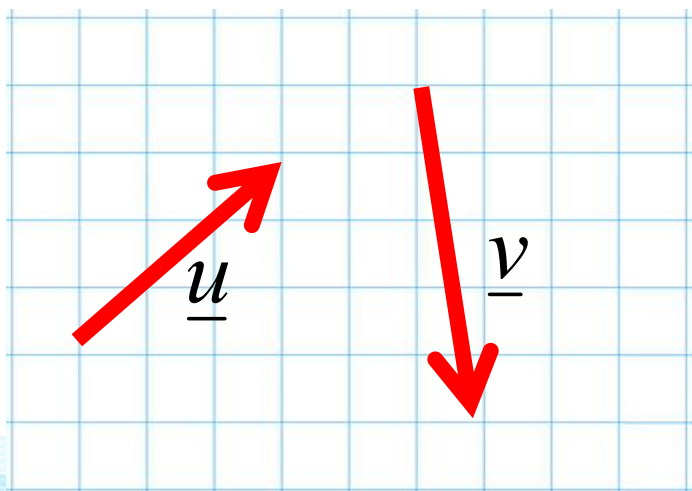
$$\therefore \overrightarrow{AB} = \frac{1}{1.5} \overrightarrow{YZ} \quad \therefore \overrightarrow{AB} \parallel \overrightarrow{YZ} \quad \text{三点不共线}$$

延申: i) 若 $m\underline{u} = n\underline{v}$, 且 $m, n \neq 0$, 则 $\underline{u}, \underline{v}$ 为平行向量。

ii) 若 $m\underline{u} = n\underline{v}$, 且 $\underline{u}, \underline{v}$ 不为平行向量, 则 $m = n = 0$ 。

iii) 若 $m\underline{u} + n\underline{v} = k\underline{u} + l\underline{v}$, 且 $\underline{u}, \underline{v}$ 不为平行向量, 则 $m = k, n = l$ 。

g) 已知 $\underline{u}, \underline{v}$, 如图, 若 $(m-3)\underline{u} = (n+2)\underline{v}$, 求 m, n 。



$$\because \underline{u} \not\parallel \underline{v}, \text{ 且 } (m-3)\underline{u} = (n+2)\underline{v}$$

$$\therefore \begin{cases} m-3=0 \\ n+2=0 \end{cases} \Rightarrow \begin{cases} m=3 \\ n=-2 \end{cases}$$

h)已知 $\underline{u}, \underline{v}$ 不为平行向量, 若 $9\underline{u} + (3h + 5)\underline{v} = (2k - 1)\underline{u} + 8\underline{v}$, 求 h, k 。

$$\because \underline{u} \not\parallel \underline{v}, \text{ 且 } 9\underline{u} + (3h + 5)\underline{v} = (2k - 1)\underline{u} + 8\underline{v}$$

$$\therefore \begin{cases} 9 = 2k - 1 \\ 3h + 5 = 8 \end{cases} \Rightarrow \begin{cases} k = 5 \\ h = 1 \end{cases}$$

i) Given that $\overrightarrow{OK} = \underline{b}$, $\overrightarrow{OL} = 2\underline{a} - \underline{b}$, $\overrightarrow{OJ} = 4\underline{a} - 3\underline{b}$.

Show that K, L, J are collinear point.

$$\overrightarrow{KL} = \overrightarrow{KO} + \overrightarrow{OL}$$

$$= -\underline{b} + (2\underline{a} - \underline{b})$$

$$= 2(\underline{a} - \underline{b})$$

$$\overrightarrow{LJ} = \overrightarrow{LO} + \overrightarrow{OJ}$$

$$= -(2\underline{a} - \underline{b}) + (4\underline{a} - 3\underline{b})$$

$$= 2(\underline{a} - \underline{b})$$

$$\because \overrightarrow{KL} = \overrightarrow{LJ}$$

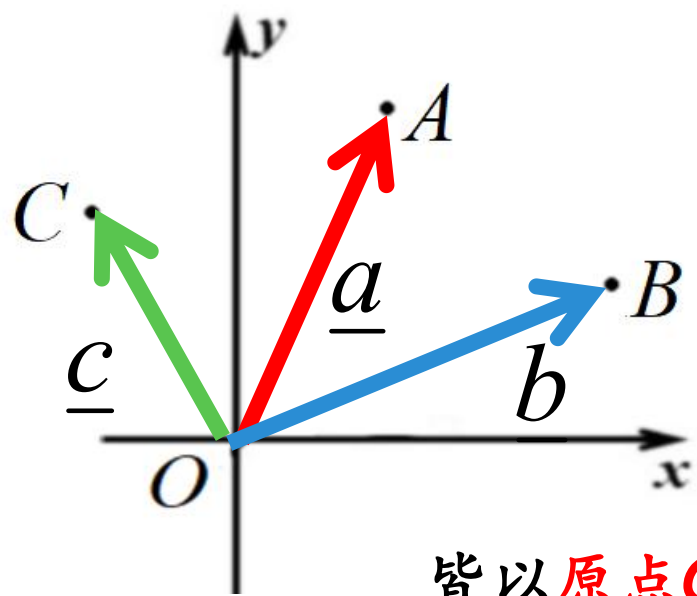
$\therefore K, L, J$ are collinear

Chp17.2 直角坐标系中的向量

位置向量

Pg. 101

位置向量：起点在**原点**的向量



$$\overrightarrow{OA} = \underline{a}$$

$$\overrightarrow{OB} = \underline{b}$$

$$\overrightarrow{OC} = \underline{c}$$

习惯以位置向量的
终点的小写字母
作为符号

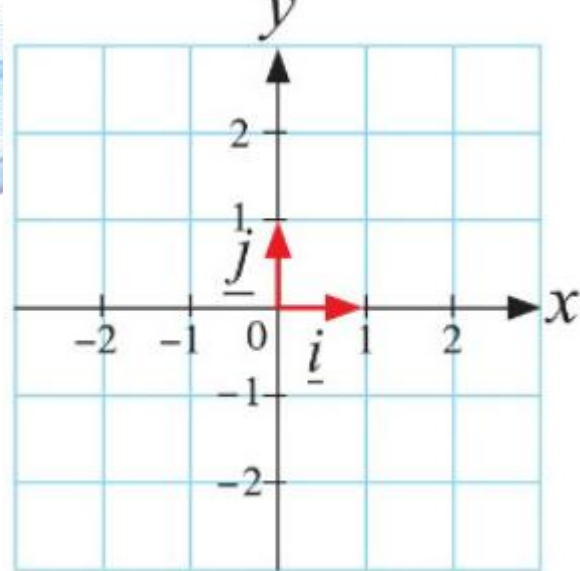
皆以**原点O**为起点

的位置向量

注：任何向量皆可以**位置向量**表示 例子： $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$

在直角坐标系中，有两个特殊的单位向量，
分别是指向正 x 轴及正 y 轴的单位向量，

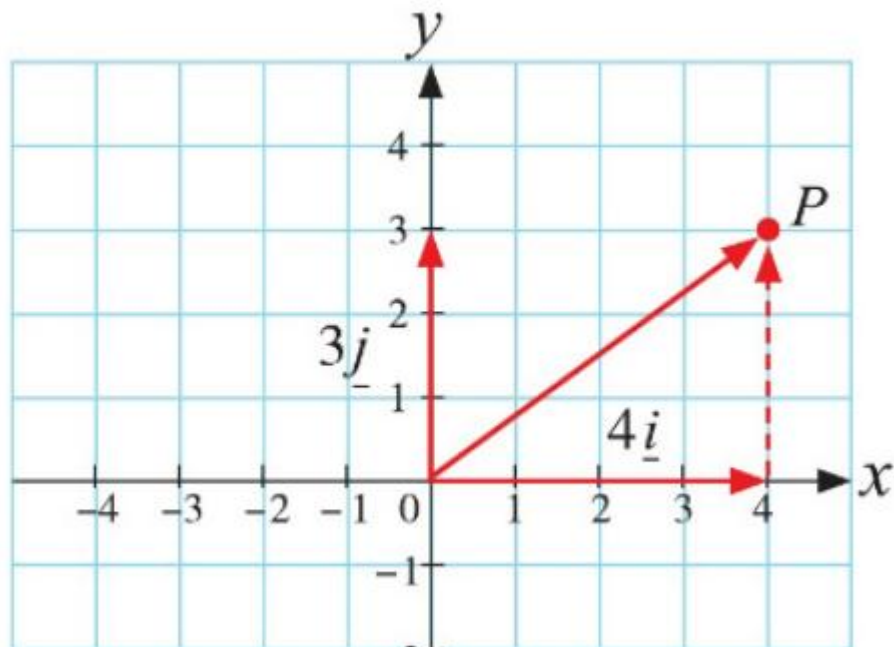
即 $\underline{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\underline{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, 见图



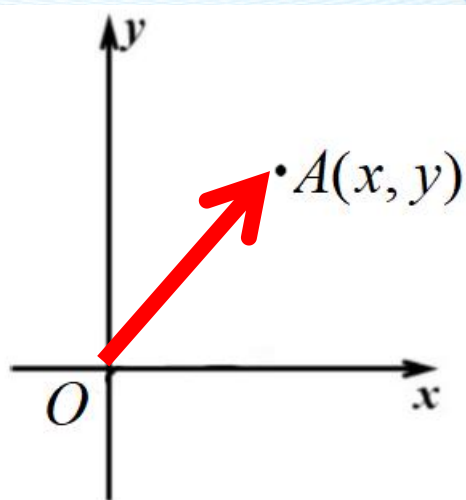
任何向量 $\underline{a} = \begin{pmatrix} x \\ y \end{pmatrix}$ 皆可表示成，

$$\underline{a} = \begin{pmatrix} x \\ y \end{pmatrix} = x\underline{i} + y\underline{j}$$

例: $\underline{a} = \begin{pmatrix} 4 \\ 3 \end{pmatrix} = 4\underline{i} + 3\underline{j}$



常用



	坐标表示法	$\underline{i}, \underline{j}$ 表示法
向量 $\overrightarrow{OA}$	$\overrightarrow{OA} = \underline{a} = \begin{pmatrix} x \\ y \end{pmatrix}$	$\overrightarrow{OA} = \underline{a} = x\underline{i} + y\underline{j}$
向量加减法、数乘	已知$\overrightarrow{OA} = \underline{a} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \overrightarrow{OB} = \underline{b} = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$ $\overrightarrow{OA} \pm \overrightarrow{OB} = \underline{a} \pm \underline{b}$ $= \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \pm \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$ $= \begin{pmatrix} x_1 \pm x_2 \\ y_1 \pm y_2 \end{pmatrix}$ $k\overrightarrow{OA} = k \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ $= \begin{pmatrix} kx_1 \\ ky_1 \end{pmatrix}$	已知$\overrightarrow{OA} = \underline{a} = x_1\underline{i} + y_1\underline{j}, \overrightarrow{OB} = \underline{b} = x_2\underline{i} + y_2\underline{j}$ $\overrightarrow{OA} \pm \overrightarrow{OB} = (x_1\underline{i} + y_1\underline{j}) \pm (x_2\underline{i} + y_2\underline{j})$ $= (x_1 \pm x_2)\underline{i} + (y_1 \pm y_2)\underline{j}$ $k\overrightarrow{OA} = k(x_1\underline{i} + y_1\underline{j})$ $= (kx_1)\underline{i} + (ky_1)\underline{j}$

a) 已知坐标 $A(3, 0)$, 坐标 $B(0, 5)$, $\overrightarrow{BC} = \begin{pmatrix} 5 \\ -6 \end{pmatrix}$, 求

i) $\overrightarrow{AB}$; ii) $\overrightarrow{AC}$; iii) C 点坐标。

$$i) \overrightarrow{OA} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}, \overrightarrow{OB} = \begin{pmatrix} 0 \\ 5 \end{pmatrix}$$

$$\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}$$

$$= -\overrightarrow{OA} + \overrightarrow{OB}$$

$$= -\begin{pmatrix} 3 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 5 \end{pmatrix}$$

$$= \begin{pmatrix} -3 \\ 5 \end{pmatrix}$$

$$ii) \overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$$

$$= \begin{pmatrix} -3 \\ 5 \end{pmatrix} + \begin{pmatrix} 5 \\ -6 \end{pmatrix}$$

$$= \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

$$iii) \overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC}$$

$$= \begin{pmatrix} 3 \\ 0 \end{pmatrix} + \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

$$= \begin{pmatrix} 5 \\ -1 \end{pmatrix}$$

$\therefore C$ 点坐标为 $(5, -1)$

b)如图, 已知坐标 $A(4,5)$, 坐标 $B(7,2)$, 且 $\overrightarrow{AB} = 2\overrightarrow{MB}$,

求 M 点坐标。

$$\overrightarrow{OA} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}, \overrightarrow{OB} = \begin{pmatrix} 7 \\ 2 \end{pmatrix}$$

$$\overrightarrow{AB} = 2\overrightarrow{MB}$$

$$\overrightarrow{AO} + \overrightarrow{OB} = 2(\overrightarrow{MO} + \overrightarrow{OB})$$

$$\overrightarrow{AO} - \overrightarrow{OB} = 2\overrightarrow{MO}$$

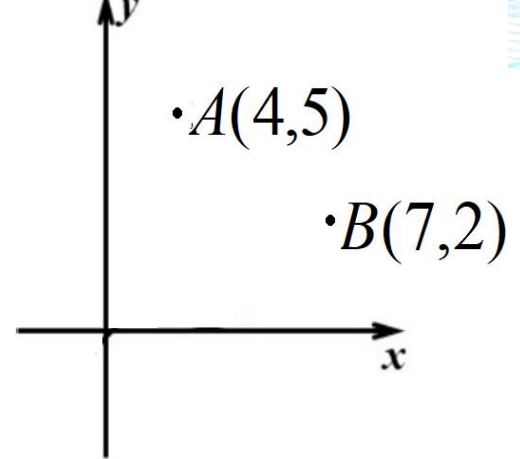
$$-\begin{pmatrix} 4 \\ 5 \end{pmatrix} - \begin{pmatrix} 7 \\ 2 \end{pmatrix} = 2\overrightarrow{MO}$$

$$-\begin{pmatrix} -11 \\ -7 \end{pmatrix} = 2\overrightarrow{MO}$$

$$2\overrightarrow{OM} = \begin{pmatrix} 11 \\ 7 \end{pmatrix}$$

$$\overrightarrow{OM} = \begin{pmatrix} \frac{11}{2} \\ \frac{7}{2} \end{pmatrix}$$

$$\therefore M \text{点坐标} \left(\frac{11}{2}, \frac{7}{2} \right)$$



c)已知 $\underline{a} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$, 求

i) $\underline{a} - \underline{b}$; ii) 若 $k\underline{a} + l\underline{b} = \begin{pmatrix} 0 \\ 7 \end{pmatrix}$, 其中 k, l 为常数, 求 k, l 的值。

$$\begin{aligned} i) \underline{a} - \underline{b} &= \begin{pmatrix} 3 \\ 4 \end{pmatrix} - \begin{pmatrix} -1 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 4 \\ 3 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} ii) \quad k\underline{a} + l\underline{b} &= \begin{pmatrix} 0 \\ 7 \end{pmatrix} \\ k \begin{pmatrix} 3 \\ 4 \end{pmatrix} + l \begin{pmatrix} -1 \\ 1 \end{pmatrix} &= \begin{pmatrix} 0 \\ 7 \end{pmatrix} \end{aligned}$$

$$\begin{cases} 3k - l = 0 & \text{--- (1)} \\ 4k + l = 7 & \text{--- (2)} \end{cases}$$

$$(1) + (2), 7k = 7$$

$$k = 1 \text{ 代入 (1), } l = 3$$

$$\therefore k = 1, l = 3$$

练习17.2(1) -3,4,7

d)已知 $\underline{a} = \begin{pmatrix} 7 \\ 6 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 2 \\ m \end{pmatrix}$, 且 $\underline{a}$ 与 $\underline{b}$ 平行, 求 m 的值。

$$\because \underline{a} // \underline{b}$$

$$\therefore \underline{a} = k\underline{b}, k \in \text{常数}$$

$$\begin{pmatrix} 7 \\ 6 \end{pmatrix} = k \begin{pmatrix} 2 \\ m \end{pmatrix}$$

$$\begin{cases} 7 = 2k & \text{--- (1)} \\ 6 = km & \text{--- (2)} \end{cases}$$

$$\frac{(1)}{(2)}, \frac{7}{6} = \frac{2k}{km}$$

$$\therefore m = \frac{12}{7}$$

e) 已知 A, B, C 的位置向量分别为 $-\underline{i} + \underline{j}$, $5\underline{i} + 7\underline{j}$, $4\underline{i} + 4\underline{j}$,

证明 $\overrightarrow{AB} \parallel \overrightarrow{OC}$ 。

$$\overrightarrow{OA} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \overrightarrow{OB} = \begin{pmatrix} 5 \\ 7 \end{pmatrix}, \overrightarrow{OC} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$$

$$\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}$$

$$= -\overrightarrow{OA} + \overrightarrow{OB}$$

$$= -\begin{pmatrix} -1 \\ 1 \end{pmatrix} + \begin{pmatrix} 5 \\ 7 \end{pmatrix}$$

$$= \begin{pmatrix} 6 \\ 6 \end{pmatrix}$$

$$\therefore \overrightarrow{AB} = \frac{6}{4} \overrightarrow{OC}$$

$$\therefore \overrightarrow{AB} \parallel \overrightarrow{OC} \quad (\text{得证})$$

f) 已知 $\overrightarrow{OA} = \begin{pmatrix} 8 \\ m \end{pmatrix}$, $\overrightarrow{OB} = \begin{pmatrix} 9 \\ 2 \end{pmatrix}$, 且 O 、 A 、 B 三点共线, 求 m 的值。

$\because O, A, B$ 三点共线

$\therefore \overrightarrow{OA} // \overrightarrow{OB}$

$\overrightarrow{OA} = k\overrightarrow{OB}, k \in \text{常数}$

$$\begin{pmatrix} 8 \\ m \end{pmatrix} = k \begin{pmatrix} 9 \\ 2 \end{pmatrix}$$

$$\begin{cases} 8 = 9k & \text{--- (1)} \\ m = 2k & \text{--- (2)} \end{cases}$$

$$\frac{(1)}{(2)}, \frac{8}{m} = \frac{9k}{2k}$$

$$\therefore m = \frac{16}{9}$$

练习17.2(2) -8,11

向量的大小

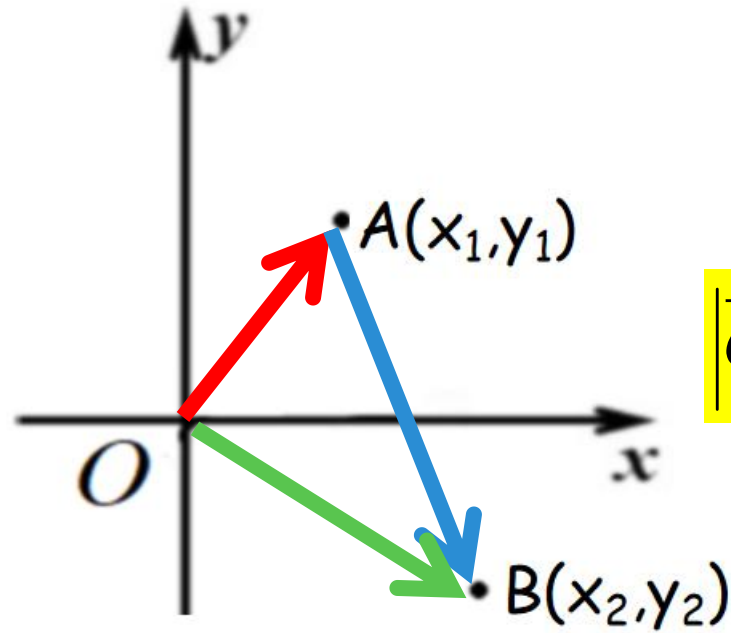
向量 $k\underline{a}$ 是向量 $\underline{a}$ 的 k 倍:

$$|k\underline{a}| = |k||\underline{a}|$$

注: $|k|$ 表示实数 k 的绝对值,

向量 $\underline{a}$ 的有向线段的长度写成 $|\underline{a}|$

例: $A(x_1, y_1), B(x_2, y_2)$



$$\overrightarrow{OA} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

$$|\overrightarrow{OA}| = \sqrt{x_1^2 + y_1^2}$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} \quad |\underline{a}| \text{ 表示向量 } \underline{a} \text{ 的大小}$$

$$= \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} - \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

$$= \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix}$$

$$|\overrightarrow{AB}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

单位向量 $\underline{\hat{a}}$:

与向量 $\underline{a}$ 具有相同方向的单位向量

$$\underline{\hat{a}} = \frac{1}{|\underline{a}|} \underline{a}$$

g) Given $\overrightarrow{OA} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$, where O is origin, find

i) coordinate of A ; ii) magnitude of $\overrightarrow{OA}$;

iii) the unit vector in the direction of $\overrightarrow{OA}$.

i) $A(2, 3)$

ii) $|\overrightarrow{OA}| = \sqrt{2^2 + 3^2}$
 $= \sqrt{13}$

iii) unit vector in the direction of $\overrightarrow{OA} = \frac{1}{\sqrt{13}} \begin{pmatrix} 2 \\ 3 \end{pmatrix}$
 $= \begin{pmatrix} \frac{2}{\sqrt{13}} \\ \frac{3}{\sqrt{13}} \end{pmatrix}$

h)已知 $\overrightarrow{AB} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$, $\overrightarrow{BC} = \begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix}$, $\overrightarrow{CD} = \frac{4p}{3} \begin{pmatrix} 2 \\ -3 \end{pmatrix}$,

i)若 $ABCD$ 为平行四边形, 求 p 的值;

ii)以 $\underline{i}, \underline{j}$ 表示 $\overrightarrow{DB}$ 。

i) $\because ABCD$ 为平行四边形

$\therefore \overrightarrow{AB} = \overrightarrow{DC}$ (同方向)

$$\begin{pmatrix} 2 \\ -3 \end{pmatrix} = -\frac{4p}{3} \begin{pmatrix} 2 \\ -3 \end{pmatrix}$$

$$2 = -\frac{8p}{3}$$

$$p = -\frac{3}{4}$$

$$ii) \overrightarrow{DB} = \overrightarrow{DC} + \overrightarrow{CB}$$

$$= \overrightarrow{AB} - \overrightarrow{BC}$$

$$= \begin{pmatrix} 2 \\ -3 \end{pmatrix} - \begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{3}{2} \\ -4 \end{pmatrix}$$

$$\therefore \overrightarrow{DB} = \frac{3}{2}\underline{i} - 4\underline{j}$$

i) Given $\overrightarrow{OA} = \begin{pmatrix} -2 \\ 6 \end{pmatrix}$, $\overrightarrow{AB} = \begin{pmatrix} 0 \\ 4 \end{pmatrix}$, C is on AB extended, $\overrightarrow{OC} = \begin{pmatrix} 4t+1 \\ -t \end{pmatrix}$, find

i) t ;

ii) $\overrightarrow{OB}$, $\overrightarrow{BC}$;

iii) coordinate of C .

iv) magnitude of $\overrightarrow{OC}$.

v) the unit vector in the direction of $\overrightarrow{OC}$.

i) $\because C$ 是 AB 延长线上

$$\therefore \overrightarrow{AC} \parallel \overrightarrow{AB}$$

$$\overrightarrow{AC} = k \overrightarrow{AB}, k \in \text{常数}$$

$$\overrightarrow{AO} + \overrightarrow{OC} = k \overrightarrow{AB}$$

$$\left[-\begin{pmatrix} -2 \\ 6 \end{pmatrix} + \begin{pmatrix} 4t+1 \\ -t \end{pmatrix} \right] = k \begin{pmatrix} 0 \\ 4 \end{pmatrix}$$

$$\begin{cases} 2 + 4t + 1 = 0 & \text{--- (1)} \\ -6 - t = 4k & \text{--- (2)} \end{cases}$$

$$\text{由 (1), } t = -\frac{3}{4}$$

$$\begin{aligned} \text{ii) } \overrightarrow{OB} &= \overrightarrow{OA} + \overrightarrow{AB} \\ &= \begin{pmatrix} -2 \\ 6 \end{pmatrix} + \begin{pmatrix} 0 \\ 4 \end{pmatrix} \\ &= \begin{pmatrix} -2 \\ 10 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \overrightarrow{BC} &= \overrightarrow{BO} + \overrightarrow{OC} \\ &= -\begin{pmatrix} -2 \\ 10 \end{pmatrix} + \begin{pmatrix} 4\left(-\frac{3}{4}\right) + 1 \\ -\left(-\frac{3}{4}\right) \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ -\frac{37}{4} \end{pmatrix} \end{aligned}$$

i) Given $\overrightarrow{OA} = \begin{pmatrix} -2 \\ 6 \end{pmatrix}$, $\overrightarrow{AB} = \begin{pmatrix} 0 \\ 4 \end{pmatrix}$, C is on AB extended, $\overrightarrow{OC} = \begin{pmatrix} 4t+1 \\ -t \end{pmatrix}$, find

i) t ;

ii) $\overrightarrow{OB}$, $\overrightarrow{BC}$;

iii) coordinate of C .

iv) magnitude of $\overrightarrow{OC}$.

v) the unit vector in the direction of $\overrightarrow{OC}$.

$$\text{iii) } \overrightarrow{OC} = \begin{pmatrix} 4\left(-\frac{3}{4}\right) + 1 \\ -\left(-\frac{3}{4}\right) \end{pmatrix} = \begin{pmatrix} -2 \\ \frac{3}{4} \end{pmatrix}$$
$$\therefore C\left(-2, \frac{3}{4}\right)$$

$$\text{iv) } |\overrightarrow{OC}| = \sqrt{(-2)^2 + \left(\frac{3}{4}\right)^2}$$
$$= \frac{\sqrt{73}}{4}$$

$$\text{iv) unit vector in the direction of } \overrightarrow{OC} = \frac{1}{\sqrt{73}} \begin{pmatrix} -2 \\ \frac{3}{4} \end{pmatrix}$$
$$= \begin{pmatrix} \frac{-8}{\sqrt{73}} \\ \frac{3}{\sqrt{73}} \end{pmatrix}$$

j)已知 $\overrightarrow{AB} = 3\underline{i} - 4\underline{j}$, $\overrightarrow{AC} = -5\underline{i} - 12\underline{j}$, 求 $\triangle ABC$ 的周长。

$$\overrightarrow{AB} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}, \overrightarrow{AC} = \begin{pmatrix} -5 \\ -12 \end{pmatrix}$$

$$\text{则 } \overrightarrow{BC} = \overrightarrow{BA} + \overrightarrow{AC}$$

$$= \begin{pmatrix} -3 \\ 4 \end{pmatrix} + \begin{pmatrix} -5 \\ -12 \end{pmatrix}$$

$$= \begin{pmatrix} -8 \\ -8 \end{pmatrix}$$

$$\begin{aligned} \triangle ABC \text{ 周长} &= \sqrt{3^2 + 4^2} + \sqrt{5^2 + 12^2} + \sqrt{8^2 + 8^2} \\ &= 18 + 8\sqrt{2} \end{aligned}$$

练习17.2(3) -6,9,12

k) 若 $\underline{a} = 4\underline{i} + 3\underline{j}$, 求

i) 与 $\underline{a}$ 同方向的单位向量;

ii) 与 $\underline{a}$ 方向相反, 大小等于15的向量 $\underline{b}$;

iii) 与 $\overrightarrow{AB}$ 方向相反的单位向量 $\underline{c}$ 。

$$i) |\underline{a}| = \sqrt{4^2 + 3^2} = 5$$

$$\text{与 } \underline{a} \text{ 同方向的单位向量 } \underline{\hat{a}} = \frac{\underline{a}}{|\underline{a}|} = \frac{4\underline{i} + 3\underline{j}}{5} = \frac{4}{5}\underline{i} + \frac{3}{5}\underline{j}$$

$$\begin{aligned} ii) \underline{b} &= -15\underline{\hat{a}} \\ &= -15\left(\frac{4}{5}\underline{i} + \frac{3}{5}\underline{j}\right) \\ &= -12\underline{i} - 9\underline{j} \end{aligned}$$

$$iii) \overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}$$

$$= -(4\underline{i} + 3\underline{j}) + (-12\underline{i} - 9\underline{j})$$

$$= -16\underline{i} - 12\underline{j}$$

$$|\overrightarrow{AB}| = \sqrt{16^2 + 12^2}$$

$$= 20$$

与 $\overrightarrow{AB}$ 方向相反的单位向量 $\underline{c}$

$$= -\left(\frac{-16\underline{i} - 12\underline{j}}{20}\right)$$

$$= \frac{4}{5}\underline{i} + \frac{3}{5}\underline{j}$$

1) 已知 $\underline{a} = 2\underline{i} - \underline{j}$, $\underline{b} = \underline{i} + 5\underline{j}$, 求 $|\underline{a} + \underline{b}|$ 及 $|\underline{a} - \underline{b}|$ 。

$$\underline{a} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \underline{b} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$$

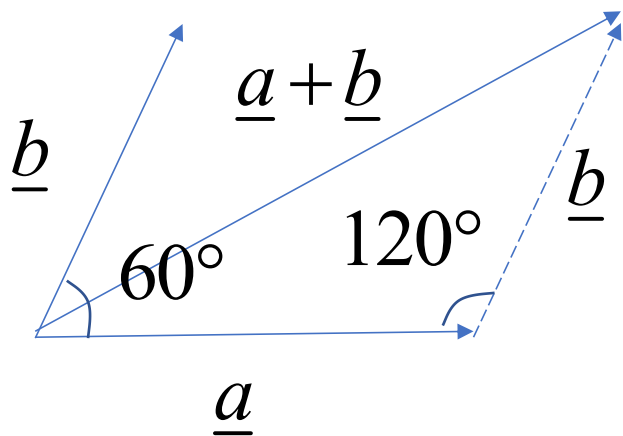
$$\underline{a} + \underline{b} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} + \begin{pmatrix} 1 \\ 5 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

$$|\underline{a} + \underline{b}| = \sqrt{3^2 + 4^2} = 5$$

$$\underline{a} - \underline{b} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} - \begin{pmatrix} 1 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \\ -6 \end{pmatrix}$$

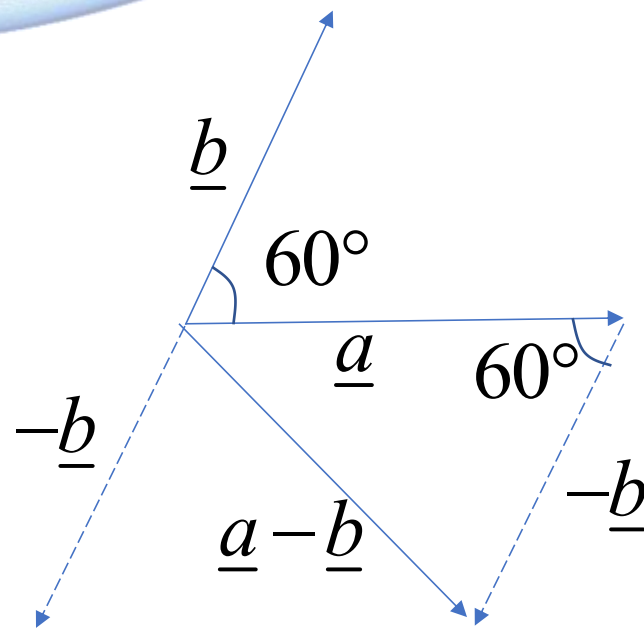
$$|\underline{a} - \underline{b}| = \sqrt{1^2 + 6^2} = \sqrt{37}$$

m) 已知 $\underline{a}$ 与 $\underline{b}$ 的夹角为 60° , 且 $|\underline{a}| = 6, |\underline{b}| = 6$, 求 $|\underline{a} + \underline{b}|$ 及 $|\underline{a} - \underline{b}|$ 。



$$|\underline{a} + \underline{b}|^2 = |\underline{a}|^2 + |\underline{b}|^2 - 2|\underline{a}||\underline{b}|\cos 120^\circ$$

$$\begin{aligned} |\underline{a} + \underline{b}| &= \sqrt{6^2 + 6^2 - 2(6)(6)\left(-\frac{1}{2}\right)} \\ &= 6\sqrt{3} \end{aligned}$$



$$|\underline{a} - \underline{b}|^2 = |\underline{a}|^2 + |\underline{b}|^2 - 2|\underline{a}||\underline{b}|\cos 60^\circ$$

$$\begin{aligned} |\underline{a} - \underline{b}| &= \sqrt{6^2 + 6^2 - 2(6)(6)\left(\frac{1}{2}\right)} \\ &= 6 \end{aligned}$$

练习17.2(4) -13,14

Chp17.3 平面向量的简单应用

Pg. 109

a) 已知坐标平面上的三点 $A(1,1)$, $B(3,2)$, $C(4,5)$, 且 $ABCD$ 为平行四边形, 求

i) $\overrightarrow{AB}$, $\overrightarrow{BC}$;

ii) D 点坐标。

$$i) \overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB} = \begin{pmatrix} -1 \\ -1 \end{pmatrix} + \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\overrightarrow{BC} = \overrightarrow{BO} + \overrightarrow{OC} = \begin{pmatrix} -3 \\ -2 \end{pmatrix} + \begin{pmatrix} 4 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

ii) 设 $D(x, y)$

$$\because \overrightarrow{AD} = \overrightarrow{BC}$$

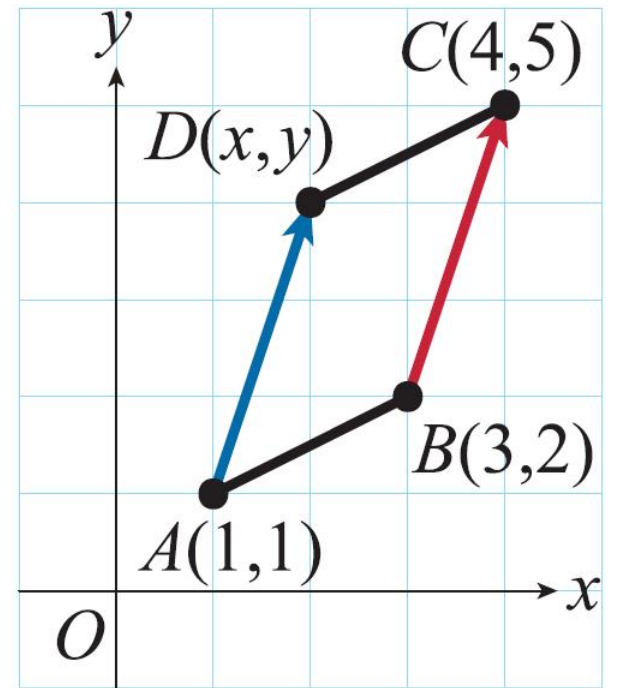
$$\therefore \overrightarrow{AO} + \overrightarrow{OD} = \overrightarrow{BC}$$

$$\begin{pmatrix} -1 \\ -1 \end{pmatrix} + \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

$$\therefore D(2, 4)$$

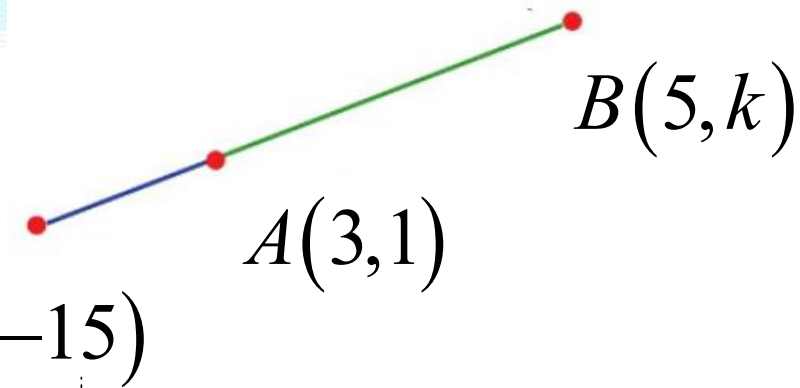
解2: AC 中点 = BD 中点



b) 已知 $A(3,1), B(5,k), C(-1,-15)$ 三点共线, 求

i) k 的值;

ii) $AB : BC$ 。



$$i) \overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB} = \begin{pmatrix} -3 \\ -1 \end{pmatrix} + \begin{pmatrix} 5 \\ k \end{pmatrix} = \begin{pmatrix} 2 \\ k-1 \end{pmatrix}$$

$$\overrightarrow{AC} = \overrightarrow{AO} + \overrightarrow{OC} = \begin{pmatrix} -3 \\ -1 \end{pmatrix} + \begin{pmatrix} -1 \\ -15 \end{pmatrix} = \begin{pmatrix} -4 \\ -16 \end{pmatrix}$$

$\therefore$ 三点共线

$$\therefore \overrightarrow{AB} = l \overrightarrow{AC}$$

$$\begin{pmatrix} 2 \\ k-1 \end{pmatrix} = l \begin{pmatrix} -4 \\ -16 \end{pmatrix}$$

$$\begin{cases} 2 = -4l & \text{--- (1)} \\ k-1 = -16l & \text{--- (2)} \end{cases}$$

$$\frac{(2)}{(1)}, \frac{k-1}{2} = \frac{-16l}{-4l}$$

$$k = 9$$

$$ii) \frac{AB}{BC} = \left| \frac{5-3}{-1-5} \right| = \frac{1}{3}$$

$$\therefore AB : BC = 1 : 3$$

练习17.3(1) -2,5,6

c)右图中, CFE 为一直线, 已知 $\overrightarrow{AB} = 20\underline{x}$, $\overrightarrow{AE} = 8\underline{y}$,

$$\overrightarrow{DC} = 25\underline{x} - 24\underline{y}, \quad AE = \frac{1}{4}AD, \quad EF = \frac{3}{5}EC,$$

i)求以 $\underline{x}$, $\underline{y}$ 表示 $\overrightarrow{BD}$, $\overrightarrow{EC}$;

ii)证明 B, F, D 三点共线。

$$i) \overrightarrow{BD} = \overrightarrow{BA} + \overrightarrow{AD}$$

$$= -20\underline{x} + 4(8\underline{y})$$

$$= -20\underline{x} + 32\underline{y}$$

$$\overrightarrow{EC} = \overrightarrow{ED} + \overrightarrow{DC}$$

$$= 3(8\underline{y}) + (25\underline{x} - 24\underline{y})$$

$$= 25\underline{x}$$

$$ii) BF = BA + AE + EF$$

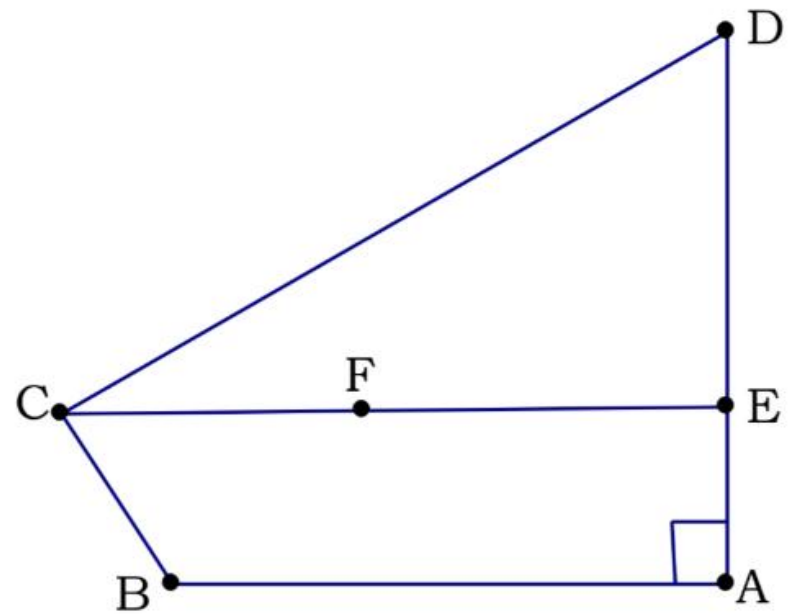
$$= -20\underline{x} + 8\underline{y} + \frac{3}{5}(25\underline{x})$$

$$= -5\underline{x} + 8\underline{y}$$

$$\therefore \overrightarrow{BD} = -20\underline{x} + 32\underline{y}$$

$$= 4\overrightarrow{BF}$$

$\therefore B, F, D$ 三点共线, 且 B 为公共点



d)如图, $\triangle OAB$ 中, P 为 AB 上一点, 且 $AP:PB=3:2$,

已知 $\overrightarrow{OP} = m\overrightarrow{OA} + n\overrightarrow{OB}$, 求常数 m, n 的值。

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP}$$

$$= \overrightarrow{OA} + \frac{3}{5}\overrightarrow{AB}$$

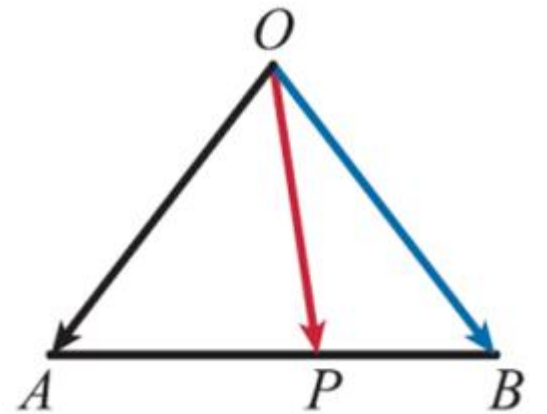
$$= \overrightarrow{OA} + \frac{3}{5}(\overrightarrow{AO} + \overrightarrow{OB})$$

$$= \overrightarrow{OA} + \frac{3}{5}(-\overrightarrow{OA} + \overrightarrow{OB})$$

$$= \frac{2}{5}\overrightarrow{OA} + \frac{3}{5}\overrightarrow{OB}$$

与 $\overrightarrow{OP} = m\overrightarrow{OA} + n\overrightarrow{OB}$ 比较系数,

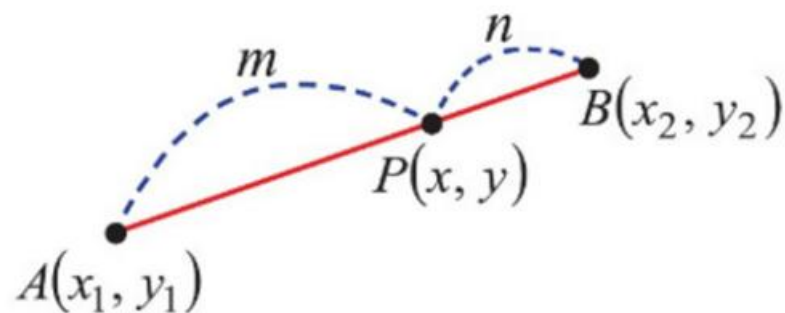
$$\text{得 } m = \frac{2}{5}, n = \frac{3}{5}$$



已知 $A(x_1, y_1), B(x_2, y_2)$ 两点, 若点 P 内分线段 AB 成 $m:n$ (如图),

则
$$\frac{AP}{PB} = \frac{m}{n}$$

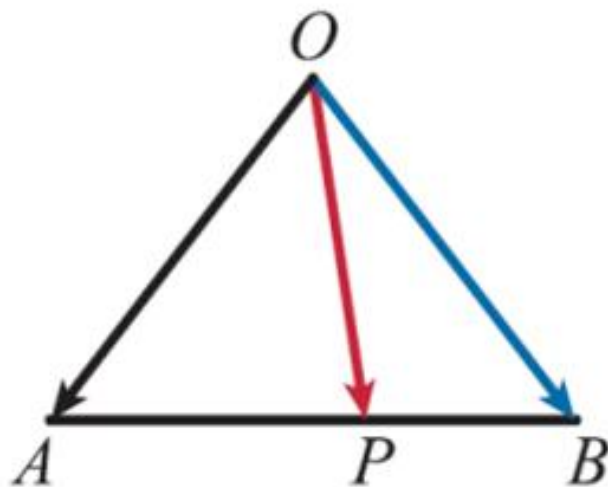
$$n\overrightarrow{AP} = m\overrightarrow{PB}$$



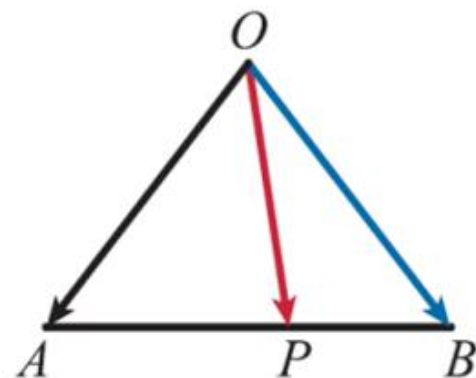
$$n(\overrightarrow{OP} - \overrightarrow{OA}) = m(\overrightarrow{OB} - \overrightarrow{OP})$$

$$(m+n)\overrightarrow{OP} = m\overrightarrow{OB} + n\overrightarrow{OA}$$

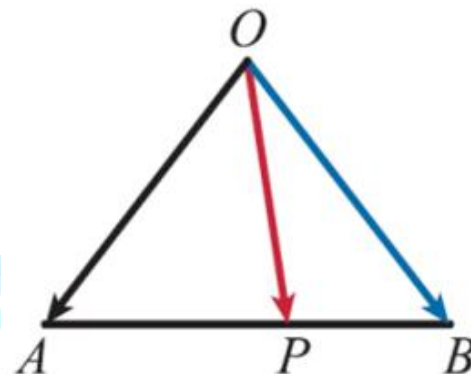
$$\overrightarrow{OP} = \frac{m\overrightarrow{OB} + n\overrightarrow{OA}}{m+n}$$



练1) $AP:PB = 5:3$,
且 $\overrightarrow{OP} = m\overrightarrow{OA} + n\overrightarrow{OB}$,
求 m, n 。



练2) $AP:PB = 5:3$,
且 $\overrightarrow{OB} = m\overrightarrow{OA} + n\overrightarrow{OP}$,
求 m, n 。



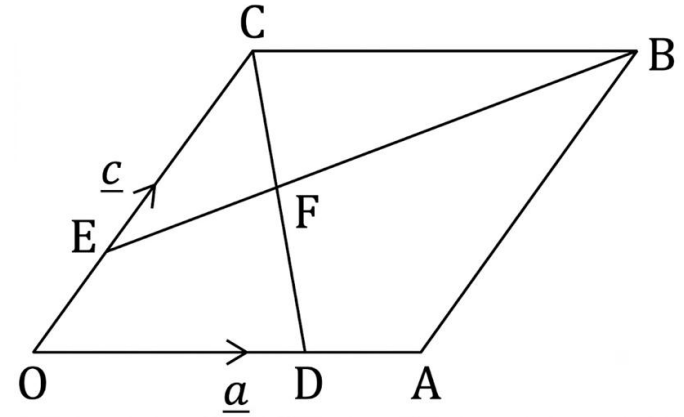
e) 右图中, $OABC$ 为平行四边形, $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OC} = \underline{c}$, $OD:DA = 3:1$,

$OE:EC = 1:2$, 直线 CD 与 BE 相交于点 F , 求

i) 若 $\overrightarrow{CF} = k\overrightarrow{CD}$, 用 $k, \underline{a}, \underline{c}$ 表示 $\overrightarrow{OF}$;

ii) 若 $\overrightarrow{EF} = l\overrightarrow{EB}$, 用 $l, \underline{a}, \underline{c}$ 表示 $\overrightarrow{OF}$;

iii) 根据 i) 及 ii), 求 k, l 的值。



$$i) \overrightarrow{OF} = \overrightarrow{OC} + \overrightarrow{CF}$$

$$= \underline{c} + k\overrightarrow{CD}$$

$$= \underline{c} + k\left(-\underline{c} + \frac{3}{4}\underline{a}\right)$$

$$= \frac{3}{4}\underline{a} + (1-k)\underline{c}$$

$$ii) \overrightarrow{OF} = \overrightarrow{OE} + \overrightarrow{EF}$$

$$= \frac{1}{3}\underline{c} + l\overrightarrow{EB}$$

$$= \frac{1}{3}\underline{c} + l\left(\frac{2}{3}\underline{c} + \underline{a}\right)$$

$$= l\underline{a} + \left(\frac{1}{3} + \frac{2}{3}l\right)\underline{c}$$

$$iii) \because \overrightarrow{OF} = \overrightarrow{OF}$$

$$\therefore \frac{3}{4}\underline{a} + (1-k)\underline{c} = \frac{1}{3}\underline{c} + l\left(\frac{2}{3}\underline{c} + \underline{a}\right)$$

$$\begin{cases} \frac{3}{4}k = l \text{——— (1)} \\ 1-k = \frac{1}{3} + \frac{2}{3}l \text{——— (2)} \end{cases}$$

$$\begin{cases} \frac{3}{4}k = l \text{——— (1)} \\ 1-k = \frac{1}{3} + \frac{2}{3}l \text{——— (2)} \end{cases}$$

$$(1) \text{ 代入 } (2), \text{ 得 } 1-k = \frac{1}{3} + \frac{2}{3}\left(\frac{3}{4}k\right)$$

$$k = \frac{4}{9} \text{ 代入 } (1), \text{ 得 } l = \frac{1}{3}$$

练习17.3(2) -8,9

f) 河水以 5ms^{-1} 从东向西流, 一船只以 6ms^{-1} 速率垂直于河流方向向北航行, 求

i) 船只实际的航行速度;

ii) 船只实际的航行速度的大小。

i) 设从东向西 $\underline{a} = \begin{pmatrix} -5 \\ 0 \end{pmatrix}$, 向北 $\underline{b} = \begin{pmatrix} 0 \\ 6 \end{pmatrix}$

$$\underline{a} + \underline{b} = \begin{pmatrix} -5 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 6 \end{pmatrix} = \begin{pmatrix} -5 \\ 6 \end{pmatrix}$$

$\therefore$ 船只实际得航行速度为 $-5\underline{i} + 6\underline{j}$

ii) $|\underline{a} + \underline{b}| = \sqrt{5^2 + 6^2} = \sqrt{61}$

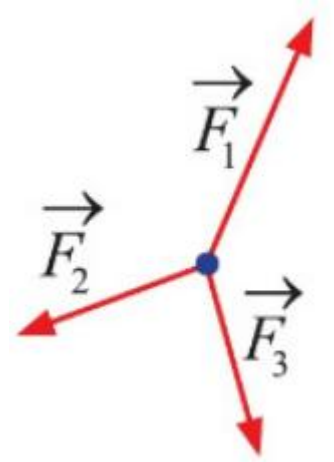
$\therefore$ 船只实际得航行速度的大小为 $\sqrt{61}\text{ms}^{-1}$

g) 右图中, 已知三个力 $\vec{F}_1 = 2\vec{i} + h\vec{j}$, $\vec{F}_2 = k\vec{i} - \vec{j}$, $\vec{F}_3 = \vec{i} - 3\vec{j}$,

同时作用于某一质点, 且物体保持平衡, 求(力的单位为牛顿N)

i) 求 h, k 的值;

ii) 若从系统中移除 $\vec{F}_3$, 求物体受到的合力大小。



$$i) \because \vec{F}_1 + \vec{F}_2 + \vec{F}_3 = \vec{0}$$

$$\therefore \begin{pmatrix} 2 \\ h \end{pmatrix} + \begin{pmatrix} k \\ -1 \end{pmatrix} + \begin{pmatrix} 1 \\ -3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{cases} 2 + k + 1 = 0 \text{———(1)} \\ h - 1 - 3 = 0 \text{———(2)} \end{cases}$$

解得, $k = -3, h = 4$

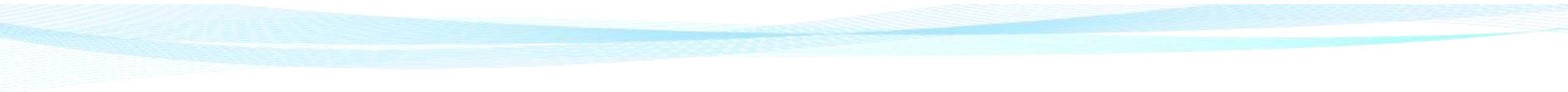
$$ii) \vec{F}_1 + \vec{F}_2 = \begin{pmatrix} 2 \\ 4 \end{pmatrix} + \begin{pmatrix} -3 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$$

$$\therefore \text{合力大小为 } \sqrt{1^2 + 3^2} = \sqrt{10}N$$

Given that $\underline{a} = p\underline{i} + 8\underline{j}$ and $\underline{b} = -3\underline{i} + 4\underline{j}$, find the value of p if

i) $\underline{a}$ and $\underline{b}$ are parallel;

ii) magnitude of $\underline{a} + \underline{b}$ is 13.

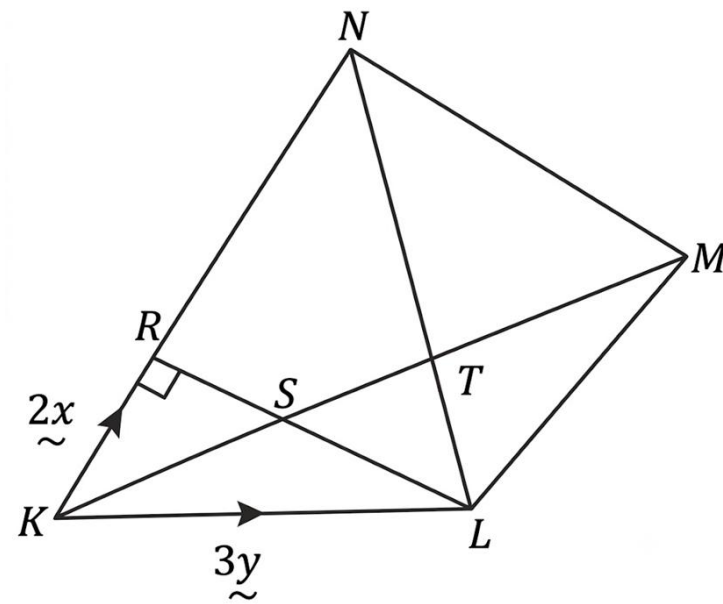


Given that $\overrightarrow{KR} = 2\underline{x}$, $\overrightarrow{KL} = 3\underline{y}$, and $LT : TN = KR : RN = 1 : 2$,

a) Express the following vectors in terms of $\underline{x}$ and $\underline{y}$.

i) $\overrightarrow{NL}$,

ii) $\overrightarrow{KT}$.



ii) Given that $\overrightarrow{NM} = -2\underline{x} + 2p\underline{y}$ and $\overrightarrow{KM} = \frac{1}{q}\overrightarrow{KT}$ where p and q are constants.

i) Express $\overrightarrow{NM}$ in terms of q , $\underline{x}$ and $\underline{y}$.

ii) Find the value of p and q .